IN-FLOW EXTRACTION OF RNA IN EXTRACELLULAR VESICLES USING A SILICON-BASED MICROFLUIDIC DEVICE

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ABSTRACT

Extracellular vesicles (EVs) transport proteins, signaling lipids, and nucleic acids from donor cells to recipient cells, therefore have been verified to serve as mediators of intercellular communications. EV-resident RNAs have been widely used as biomarkers of disease and prognostic indicators. Thus, in situ extraction of RNAs in a microfluidic flow is of great importance for developing integrated microsystems to realize on-chip PCR. This paper presents a silicon-based microfluidic device for solid-phase extraction of EV-resident RNAs. The microfluidic device contains micro-pillar arrays within an area of 0.25cm^2 on the silicon substrate, different structures with various surface-to-volume ratio (SVR) are investigated to compare their RNA extraction efficiencies. A 20% extraction efficiency is obtained and characterized by using a Nanodrop UV-Vis spectrophotometer. The proposed on-chip extraction of EV-resident RNAs would provide new approaches to develop integrated microfluidic systems to realize PCR for point-of-care applications.

KEYWORDS

EV-resident RNAs, Microfluidics, Micro-pillar array, Solid-phase extraction.

INTRODUCTION

EVs are membrane-enclosed structures that are secreted from numerous cell types. A variety of molecules have been identified inside EVs, including proteins, DNAs, RNAs, etc [1]. Since EVs are derived from intracellular material, the cargo transportation from donor cells to recipient cells could help to elucidate intercellular communication. Recent works have demonstrated the roles of EVs in many processes of physiology and pathology from molecular level to tissue level [2-3].

EV-resident RNA has gained great attention as novel meditators during intercellular communication. It has been reported that EVs are abundant in microRNAs (miRNAs), along with other types of RNAs including transfer RNAs (tRNAs), long noncoding RNAs (lncRNAs), mitochondrial RNAs and ribosomal RNAs (rRNAs) [4-5]. These RNAs are released in a variety of subpopulations from different cell lines via various pathways, which serve as regulator and translator in recipient cells. As vital information source of intra- and intercellular processes, efforts have been made by studying their biogenesis, compartmentalization and vesicle packaging. Yet the relatively low abundance of EVs and their RNAs presents quite some challenges. Although highly sensitive methods, particularly RNA-sequencing and reverse transcription and quantitative PCR analysis (RT-qPCR), afford accurate and quantitative identification of RNAs, all requiring considerably amounts of time and cumbersome protocols.

Most of current commercial RNA extraction kits use silica columns to capture RNAs, sometimes with positive amino-groups derivative to bind negatively charged RNAs more effectively. However, the above-mentioned protocols are involved with chemicals, i.e. ethanol and chaotropic salts, which could inhibit RT-PCR reaction or other molecular techniques. The required washing steps will inevitably compromise the RNA extraction yield. Therefore, the extraction of low abundant EV-resident RNAs demands a more straightforward protocol. Microfluidic Solid-Phase Extraction (SPE) methods may provide an alternative to realize RNA extraction and preconcentration from complex biological samples with sample volumes on the order of tens of microliters. More importantly, it allows more straightforward sample preparation and further RT-PCR reaction on-chip. Low concentration of cell-free DNA extraction has already been reported by SPE on silicon-based microfluidic platform and then eluted with appropriate buffers, with good enough quality and quantity for the following PCR reaction [6-7]. Conceivably, a SPE based microfluidic platform would be a plausible toolbox for low abundant EV-resident RNA extraction and following analysis.

This paper presents a microfluidic platform for EV-resident RNA extraction and potential RT-PCR on chip (Figure 1). In contrast to complicated protocols such as extraction, purification, concentration, amplification of conventional RNA analysis, this platform allows straightforward extraction/preconcentration of RNA and RT-PCR on-site. It provides an alternative approach to identify EV-resident RNAs and investigate their biological roles in regulating inter- and intracellular progress.

MATERIALS AND METHODS

Silicon-based Microdevice Fabrication

The presented microfluidic platform (15 mm × 8 mm) consists of a silicon substrate with micro-pillar features in an etching area (5 mm × 5 mm) and a thin film of Polydimethysiloxane (PDMS, Sylgard 184, Dow Corning GmbH, Wiesbaden, Germany) as the top layer to enclose the microfluidic channels. The pillars are of 100 μm in height, with varying diameters (from 10 to 20 μm) and pillar-to-pillar distances (from 10 to 20 μm) to achieve different SVR.

The silicon chip was fabricated by dry etching process. Silicon wafer (Ø4 in., 450 μm in thickness) was firstly

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coated by a thin layer (200 nm in thickness) of SiO_2 that was made by thermal oxidation and used as the hard mask for the etching processes. The hard mask was then patterned by standard photolithography using AZ4620 positive photoresist (11 µm in thickness, MicroChemicals GmbH, Ulm, Germany), followed by reactive ion etching (RIE) using CF_4 + CHF_3 plasma (RIE-10NR, Samco Inc., Kyoto, Japan) to open the window for silicon deep etching on the SiO_2 layer. Then the remaining photoresist layer was stripped, leaving the window for silicon etching fully open. Deep RIE (DRIE) with C_4F_8 + SF_6 plasma (DSE200S, NAURA Technology Group Co., Ltd., Beijing, China) was used to etch the silicon substrate for a depth of 100 µm. After wafer dicing, the fabricated silicon chips were then cleaned by piranha solution (3 H_2SO_4: 1 H_2O) and dried on a hot plate at 100 °C for 30 minutes. Finally, the chip was sealed by a PDMS slab using oxygen plasma bonding.

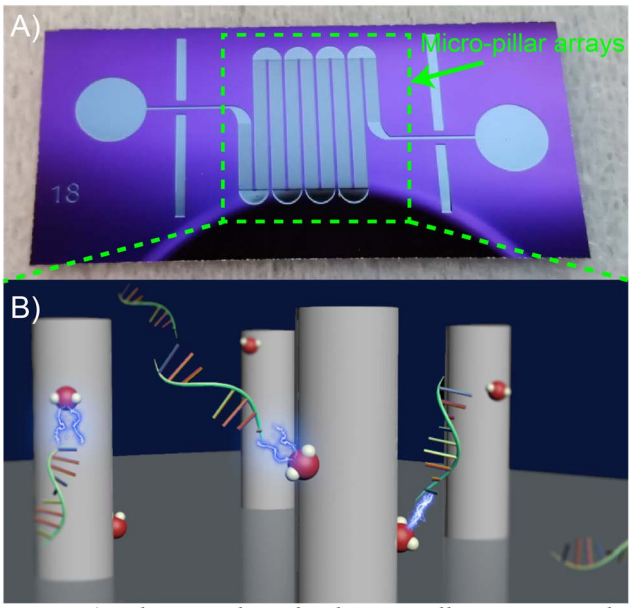

Figure 1: Photograph and schematic illustration on the microfluidic device. (A) Image of the microfluidic chip; B) Schematic of RNA extraction on chip. RNA molecules are bound to the APTES-coated surface of the micro pillars by electrostatic interaction between the positive charged amino groups on the surface and the negative charged RNA molecules.

Surface Modification

Before RNA extraction, the silicon chip needed to graft amino-group on the surface in order to improve the RNA capture efficiency. The experimental set-up required the silicon chip to be mounted on a peltier module and connected to a syringe pump (neMESYS 290N, Cetoni GmbH, Korbußen, Germany). The silanization was implemented by in-flowing 1% (v/v%) 3-aminopropyltriethoxysilane (APTES, molecular weight: 221, Sigma-Aldrich, St. Louis, MO, USA) solution inside microfluidic channels. In total 50 µL of APTES solution was injected at a flow rate of 1 µL/min on preheated silicon chip at 60°C. Then channel rinsing was carried out with 50 µL RNase-free H_2O at a flow rate of 5 µL/min to remove any unbounded APTES.

Sample Preparation

Exosomes derived from A549 human non-small cell lung cancer cells were purified from A549 cell culture medium by using Exo-Spin exosome purification kit (Cell Guidance Systems Ltd., London, UK). 30 mL of culturing medium, which was equivalent to about 1×10^8 cells, was collected after 48 hours for exosome purification. The purified exosomes were characterized by Nanoparticle Tracking Analysis (NTA, ZetaView PMX 110, Particle Metrix GmbH, Inning am Ammersee, Germany) and a flow cytometer (CytoFLEX S, Backman Coulter, Brea, USA) as demonstrated in [8]. ExoRNeasy Midi and Maxi Kits (Qiagen GmbH, Hilden, Germany) were used for total RNA extraction from the purified exosomes under operation instructions provided by the manufacturer. The resident RNAs were then eluted in RNase-free H_2O, the concentration was determined by a UV spectrophotometer (Nanodrop 2000, ThermoFisher Scientific, Waltham, USA).

On-chip RNA Extraction Experiments and Performance Characterization

To perform on-chip RNA extraction, the above-extracted RNA was introduced to the microchannel at a flow rate of 1 µL/min, with concentrations ranging from 12.5 – 175 ng/µL dissolved in 20 µL RNase-free H_2O, i.e. the total initial RNAs load was ranged between 250 – 3500 ng. Afterwards, the microchannel was rinsed with 20 µL 75% ethanol. Finally, to elute the bounded RNAs, the microchannel was flushed with 20 µL RNase-free H_2O at a flow rate of 10 µL/min. All eluents at each step were collected and analyzed by UV spectrophotometry measurements. Besides, to highlight the benefits of APTES coating, a control group with unmodified chip was also included in the study under the same protocol as above.

RESULTS

Geometrical Features of the Micro-Pillar Arrays

One major factor of RNA extraction by SPE using micro-pillar approach is the SVR: the higher the value, the larger the surface available for RNA capture given the same chemical conditions (concentration, pH, etc.). To study how the SVR relates to the extraction performance of our device, three different pillar microstructures featuring different SVR values were designed as in Table 1.

Table 1 Pillars geometries featuring different SVR values

Chip No.	Pillar diameter (µm)	Pillar distance (µm)	Pillar height (µm)	Pillar number	SVR value
1	10	10	100	41472	86.9
2	15	15	100	23904	62.8
3	20	20	100	11904	32.9

Figure 2 shows the silicon micro-pillar features in Chip No.1. The cylindrically shaped pillars are 10 µm in diameter (15 µm in the case of Chip No.2 and 20 µm for Chip No.3) and 20 µm in inter-pillar distance (15 µm in the case of Chip No.2 and 20 µm for Chip No.3). Noted that the major contribution to the total SVR value comes from

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the active pillar area. Such area takes about 83.9% for Chip No.1, 81.8% for Chip No.2 and 74.9% for Chip No.3 from the total surface area of the chip. Therefore, we can infer that the active pillar area contributes the major impact during RNA extraction.

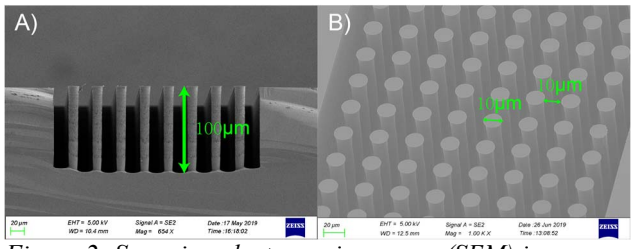

Figure 2: Scanning electron microscopy (SEM) images on the micropillars. (A) Image showing the height of the micropillars; (B) A tilted view of micro-pillar array.

EV Analysis

The size and concentration of the extracted EVs were measured by NTA. A narrow peak at 110 nm was observed, with the size distribution ranged from 70 nm to 200 nm (see Figure 3A). The concentration of the extracted EVs was approximately 2.0×10^{10} particles/mL, of which there were $77.9 \pm 2.2\%$ of EVs larger than 100 nm.

The composition of the extracted EVs was further evaluated by aptamer-facilitated dual staining method as in [8]. A Cy5.5-labeled biotinylated DNA aptamer against CD63, i.e. a transmembrane protein which enriches on EV's membrane), and Carboxyfluorescein diacetate N-succinimidyl ester (CFSE, a commonly used cytoplasmic dye) were implemented to label bioactive EVs after extraction. Flow cytometer screening indicated that up to 45.03% of the extracted substance were bioactive EVs in the 1st quadrant, while protein aggregates occupied over 33.79%, 10.38% unused labeling aptamer, and 10.8% unstained contaminants in the extracted sample (Figure 3B).

Effect of SVR on RNA Extraction Performance

The binding and elution capability of the micro-pillar chip was evaluated by UV spectrophotometry measurement of collected samples from loading, washing and elution steps. Figure 4 illustrates the influence of different SVR values on the RNA extraction profile, using amino-coated micropillars with various initial RNA loading amounts. For each data set, the non-trapped (recovered solution after initial loading), washed (washing solution) and eluted (eluent solution) RNAs were collected and plotted. Noted that the amount of bounded RNA is divided into: 1) a portion strongly bounded on the amino-coated surface which is later collected in the elution step; 2) a portion loosely attached on the surface which can be easily washed off in the washing step. It is obvious that higher extraction efficiency (i.e. amount of strongly bounded RNA / initial loading amount of RNA) can be achieved in higher SVR design, as expected on the basis of increasing potential binding surface sites upon higher SVR value. Remarkably, the extraction efficiency is hindered given higher initial loading amount. Higher percentage of non-trapped RNA is observed under initial loading which is greater than 2000 ng (See Figure 4A). This may suggest that a level of saturation was reached. Among the above-mentioned designs, the extraction efficiency can reach up to 20% in the lowest initial loading (~12.5 ng/μL initial concentration), while decrease at higher initial loading (See Figure 4F). This implicates potential advantage in the scenario of low abundant or rare RNA sample analysis. On the other hand, Figure 4E shows less than 1% extraction efficiency corresponding to uncoated micropillars, meaning that most amount of RNAs just flowed through without even loosely bounded to the surface. This also indicates that the majority of the extracted RNAs were indeed bounded to the amino-coated surface.

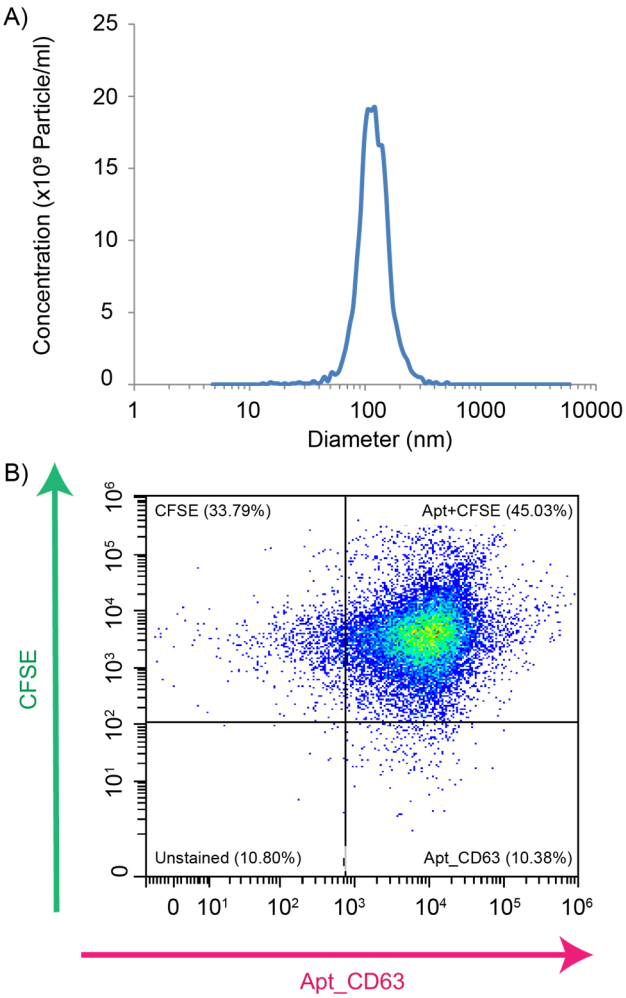

Figure 3: Characterization of the EV samples. (A) NTA profile of the extracted EVs from A549 cells; (B) EV composition analysis by CFSE & Apt_CD63 staining using high resolution flow cytometer

Ongoing PCR Experiments on the Integrated System

As a preliminary step towards the development of a miniaturized EV-resident RNA analysis system, we proposed the plan with in-situ real time amplification on the same microfluidic chip in an integrated configuration. That is, by implementing miniaturized thermal module, the microfluidic chip is to perform RT-qPCR for further transcriptomics study of EVs. Real samples could also be considered, such as spiked-in RNA serum, or body fluids like blood, urine, and saliva etc. Therefore, we propose an

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improved microfluidic platform, which will contain EV sample lysis, EV extraction, RT-qPCR on chip in a straightforward format comparing to traditional benchtop RNA analysis methods. The experiments are being performed and preliminary results verified its feasibility.

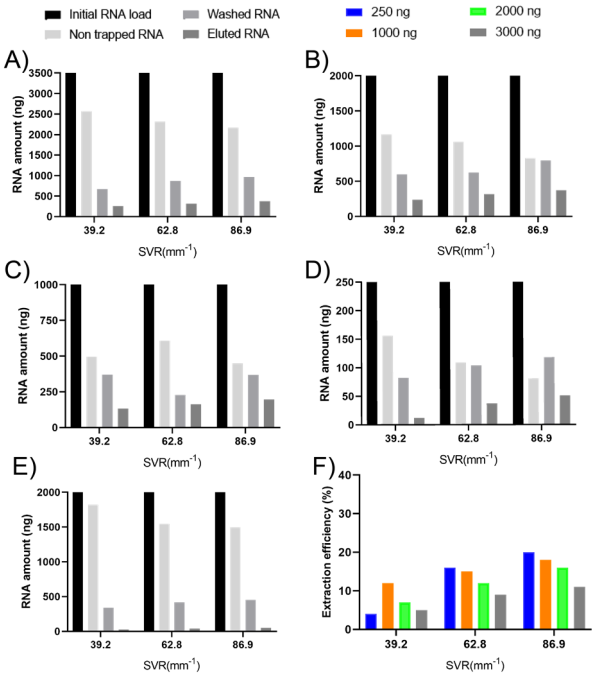

Figure 4: On-Chip RNA extraction profile as a function of SVR value corresponding to different initial RNA loading amounts of (A) 3500 ng; (B) 2000 ng; (C) 1000 ng; (D) 250ng. (E) A control group of uncoated chip using 2000 ng of initial RNA loading. (F) Extraction efficiency over various SVR value of the microfluidic chip, given various initial RNA loading amounts as indicated in the legend.

CONCLUSION

We have presented a microfluidic platform for low abundant EV-resident RNA extraction based on amino-coated silicon micropillars. Extraction performance has been demonstrated by spike-in EV-resident RNA solutions. The extraction efficiency reaches about 20% in the case of the device exhibiting the highest SRV value. Besides, higher extraction efficiency has been obtained in lower initial RNA loading amount, denoting advantages in extracting low abundant or rare EV-resident RNA samples. Preliminary results encourage us in continuing our work towards a fully integrated RNA extraction / RT-qPCR microdevice. This will facilitate transcriptomics study of EVs in a much more straightforward format for real samples analysis and point-of-care applications.

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